

Network and IoT Security | NIOsec

Summary

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1. INTRODUCTION

<i>Communication layers</i>	<i>Security protocols</i>
Application layer	Web Application Security, VoIP Security, SW Security
Transport layer	TLS
Network layer	IPsec
Data Link layer	L2TP, IEEE 802.1X, IEEE 802.1AE, IEEE 802.11i (WPA2)
Physical layer	Quantum Key Exchange

The CIA triad is a foundational information-security model stating that systems should protect:

Confidentiality: Keeping information secret

Integrity: Keeping information correct and unaltered

Availability: Ensuring information and systems remain accessible

Additional desired properties:

Authenticity: Adversary cannot claim to be someone else

Accountability/Non-repudiation: Actions can be traced to actor

TODO:

W1 slides 15,16

2. TLS

SSL and TLS are protocols for internet handshakes and encrypted transmission. Secure Socket Layer (SSL) came first, then after v3.0 it became Transport Layer Security (TLS), currently v1.3. People still use SSL as a term, even though technically it's now TLS.

TLS 1.2: Introduced authenticated encryption, which is the most modern form of encryption. It's a more flexible protocol, lots of TLS extensions, the hashing function has been improved.

TLS 1.3: The protocol has been redesigned to make it faster. A lot of the old ciphers have been removed to make it more secure. It only supports authenticated encryption (AEAD).

TODO:

W1 slides 22

2.1. SPECIFICATION

2.1.1. Record protocol

All TLS packets are sent using the record protocol. This header is attached to every single message.

TODO:

slides 23 frame((Type: 1, Version: 2, Length: 2,))

2.1.2. Subprotocols

TODO:
slides 24

2.1.3. Handshake

TODO:
slides 25-26

2.2. MODERN TLS VARIANTS

2.2.1. DTLS

TODO:

2.2.2. QUIC

TODO:

2.3. ATTACKS

2.3.1. Heartbleed

TODO:

2.3.2. Syn Flooding

TODO:

2.3.3. TCP Reset

TODO:

2.3.4. FREAK

TODO: